1. In a Wumpus World, the agent observes that there is no breeze in square [1,1]. A square has a breeze if and only if at least one of its adjacent squares contains a pit.

   LetB11$B$11: There is a breeze in square[1,1][1,1]

   P12$P$12:There is a pit in square[1,2][1,2]
   P21$P$21: There is a pit in square[2,1][2,1]
   Answer the following:
   a) Formulate the Knowledge BaseKBKBusing propositional logic.
   b) Let α1 =¬P12$P$12. Using logical reasoning, determine whether KB⊨α1 and justify your answer.
   Ans: a)A square has a breeze **if and only if** at least one adjacent square contains a pit.
   Therefore, the Wumpus World rule is: B11⇔(P12∨P21)
   KB={¬B11,B11⇔(P12∨P21)}KB={¬$B$11,$B$11⇔($P$12∨$P$21)}
   b) Given:
   α1 =¬P12
   From the KB:B11⇔(P12∨P21) herefore:
   KB⊨α1
   KB⊨¬P12

2. Define Wumpus World. List any three percepts that an agent can receive in the Wumpus World.

Ans: **Wumpus World** is a simple grid-based environment used in Artificial Intelligence to study **knowledge representation, logical reasoning, and intelligent agents**.

Any three percepts:

- **Stench** – indicates that the Wumpus is nearby.
- **Breeze** – indicates that a pit may be in an adjacent square.
- **Glitter** – indicates that gold is present in the current square.
- **Bump** – indicates that the agent has hit a wall.
- **Scream** – indicates that the Wumpus has been killed.

3. Define Propositional Logic. Explain the basic components of Propositional Logic with a suitable example.

Ans: **Propositional Logic** is a formal system in which statements are represented as propositions that are either **True or False**.

Its basic components include:

- **Propositions:** P: It is raining
- **Logical connectives:** AND (∧), OR (∨), NOT (¬), IMPLIES (→)
- **Compound propositions:** P ∧ Q, P → Q

Example:

Let P = It is raining and Q = The road is wet.

Then P → Q means: **If it is raining, then the road is wet.**

4. Consider a 4 × 4 Wumpus World environment. An intelligent agent starts at cell (1,1) and its goal is to reach the cell containing Gold without falling into a pit or being eaten by the Wumpus.

The agent can move Up, Down, Left, or Right between adjacent cells. It receives the following percepts:

- Breeze near a pit
- Stench near the Wumpus
- Glitter at the location of Gold

i) Explain the rules of the Wumpus World and the significance of the Breeze, Stench, and Glitter percepts.

ii) Determine a safe sequence of movements for the agent to travel from (1,1) to collect the Gold without falling into a pit or being eaten by the Wumpus. Show the cells visited by the agent.

iii) Explain how the agent uses its percepts and knowledge of the environment to decide which cells are safe to enter.

Ans: i)Wumpus World is a grid-based environment in which an intelligent agent must find the Gold while avoiding dangerous cells containing Pits and the Wumpus.

The important percepts are:

1. Breeze: If the agent perceives a Breeze in a cell, it indicates that a Pit is present in one of the adjacent cells.
2. Stench: If the agent perceives a Stench, it indicates that the Wumpus is present in one of the adjacent cells.
3. Glitter: If the agent perceives Glitter, it means that Gold is present in the current cell. The agent can then collect the Gold.

The agent must use these percepts to infer which cells are safe before moving.

ii) (1,1) start

percepts (none, none , none , none, none)

(1,1) to (1,2)

percept sequence (none, breeze, none, none, none)

pit may be in (3,1) or (2,2)

come back to (1,1) indicate the visited node

move from (1,1) to (1,2)

percept sequence (stench, none, none, none, none) - Wumpus can be in (3,1) or (2,2). But Wumpus is not in (2,2) . So move to (2,2). Now move to (2,3). Percept sequence (stench, breeze, glitter, none, none). Gold found. collect gold and come back to (1,1) through same path.

iii)The agent uses its percepts to determine safe movements:

- A Breeze warns the agent about a possible Pit in an adjacent cell, so it avoids entering suspicious cells.
- A Stench warns the agent about the nearby Wumpus, so it avoids the corresponding dangerous cell.
- Glitter confirms that the Gold is in the current cell, so the agent stops and collects it.

Thus, the agent combines its percepts and knowledge of previously visited cells to select a safe path and successfully reach the Gold without being killed.

Draw diagram

5. Discuss the PEAS description (Performance measure, Environment, Actuators, Sensors) for the Wumpus World agent.

   Ans:

Component	Description
Performance measure	+1000 for gold, −1000 for death, −1 per action, −10 for using the arrow
Environment	4×4 grid of rooms, pits, Wumpus, gold
Actuators	Move forward, turn left/right, grab, shoot, climb out
Sensors	Stench, breeze, glitter, bump, scream

6. Describe the Wumpus World environment with a suitable diagram. Explain its components — caves, pits, gold, and the Wumpus.

7. Differentiate between declarative and procedural approaches to building a knowledge base giving appropriate examples.

8.

Explain the architecture of a knowledge-based agent. Describe the roles of the knowledge base, TELL, and ASK operations with a neat diagram

9.

Define entailment in propositional logic with an example.

10.

Define a knowledge-based agent and describe the TELL and ASK operations used with a Knowledge Base (KB).

11.

Explain the difference between syntax and semantics in First-Order Logic with suitable examples.

12.

Explain the knowledge engineering process in First-Order Logic. Describe the major steps involved in developing a knowledge base, from identifying the task to testing the knowledge base.

13.

Write short notes on Higher-order logic and mention how it differs from First Order Logic.

14.

Explain the Wumpus World knowledge base using First Order Logic. Write axioms for perception, and how the agent infers the location of pits using FOL

Ans: The Wumpus World is a classic environment used to demonstrate knowledge representation and reasoning in FOL, where an agent must navigate a cave, avoid pits and the Wumpus, and find gold, using percepts (Stench, Breeze, Glitter) to infer the environment's hidden state.

Perception axioms:

The agent perceives Breeze, Stench, Glitter, Bump, Scream at each time step, represented as:

Percept([Stench, Breeze, Glitter, None, None], 4)

To draw conclusions from percepts, we use a **rule linking percept and property**:

$\forall b, g, u, c, t\ \text{Percept}([\text{Stench}, b, g, u, c], t) \Rightarrow \text{Stench}(t)$
$\forall s, g, u, c, t\ \text{Percept}([s, \text{Breeze}, g, u, c], t) \Rightarrow \text{Breeze}(t)$

Breeze perceived at a square implies a pit in a neighboring square:

$\forall s, t\ (\text{At}(\text{Agent}, s, t) \wedge \text{Breeze}(t)) \Rightarrow \text{Breezy}(s)$

Squares are breezy exactly when a neighboring square has a pit (biconditional, causal rule):

$\forall s\ (\text{Breezy}(s) \Leftrightarrow \exists r\ (\text{Adjacent}(r, s) \wedge \text{Pit}(r)))$

Similarly for Stench and Wumpus:

$\forall s\ (\text{Smelly}(s) \Leftrightarrow \exists r\ (\text{Adjacent}(r, s) \wedge \text{Wumpus}(r)))$

15.

Explain the Knowledge Engineering process in FOL. List and describe all the steps involved, illustrating with a suitable example domain

16.

Explain the syntax and semantics of First-Order Logic. Describe terms, atomic sentences, complex sentences and the notion of a model.

17.

Consider a FOL model with a domain of 3 objects: {a, b, c}, and a binary predicate R(x, y).

i) How many distinct ground atomic sentences can be formed using R over this domain?

ii) How many distinct possible interpretations (extensions) of R exist over this domain ?

18.

Differentiate between the Universal Quantifier (∀) and the Existential Quantifier (∃) in FOL. Explain why ∀ is conventionally paired with → while ∃ is conventionally paired with ∧, giving an example of each.

Ans: **Universal Quantifier (∀):** States that a sentence holds true for **every** object in the domain of discourse. Example: ∀x (King(x) → Person(x)) — "every king is a person"

Existential Quantifier (∃): States that there is **at least one** object in the domain for which the sentence holds true. Example: ∃x (Crown(x) ∧ OnHead(x, John)) — "there exists a crown that is on John's head"

Why ∀ pairs with → and ∃ pairs with ∧:

- With ∀, using **∧** instead of → would be a common and serious mistake: ∀x (King(x) ∧ Person(x)) would incorrectly assert that **every object in the entire domain is both a king and a person** — including rocks, numbers, etc. Using → correctly restricts the assertion: ∀x (King(x) → Person(x)) only asserts something about objects that *are* kings — for all other objects, the implication is vacuously true (harmless).
- With ∃, using → instead of **∧** would create a different problem: ∃x (King(x) → Person(x)) would be satisfied by **any** object that is *not* a king (since a false antecedent makes → vacuously true) — even if no king exists at all! Using **∧** correctly requires that the found object actually **is** a king **and** actually is a person: ∃x (King(x) ∧ Person(x)).

19.

Translate the following English sentences into First-Order Logic: (a) "All humans are mortal." (b) "Some students like Artificial Intelligence."

20.

Define First-Order Logic (FOL). List its basic syntactic elements with one example of each.